

exactly 80 mm: _____

FIRST SOLVE

New here? Do the walk-through once at cubepath3.svn.sourceforge.net/viewvc.cgi?view=rev&rev=103. This card is the memory jog.

NOTATION
R = turn the face clockwise, looking at it from outside the cube.
r = counter-clockwise. **2** = half turn. Lowercase **r** = that face plus the layer behind it, turning together.
y = spin the whole cube left, white staying down. A whole-cube turn solves nothing.

WORDS
 algorithm = a fixed sequence of turns · trigger = a short algorithm you have to run on cue · unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out. Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
R'UR'



white sticker FRONT
L'UL



white sticker UP run the first one once to knock it out, then look again
R'UR'

3 TOP EDGE SIDES

Middle edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.



goes RIGHT
U R'UR' y L'ULU



goes LEFT
U L'ULU y' R'UR'U

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

■ **RUR'** sexy move
 ■ **RUR** U Sine
 ■ **R'FRF** sledhammer
 gap = change grip

FIRST SOLVE

New here? Do the walkthrough once at cubepath-53.vercel.app/learn. This card is the memory log.

NOTATION
 R U' F D B = turn that face clockwise, looking at it from outside the cube
 r = counter-clockwise, 2 = half turn. Lowercase r f = that face plus the layer behind it, turning together
 y = spin the whole cube left, White staying down. A whole-cube turn solves nothing.

WORDS
 algorithm = a fixed sequence of turns + trigger = a short algorithm you're hands on to use on unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it. Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
R'U'R

white sticker FRONT
L'U'L

white sticker UP turn the first one once to knock it out, then look again.
R'U'R

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.

goes RIGHT
U' R'U'R' y' L'U'L

goes LEFT
U' L'U'L y' R'U'R'



Edge stuck in the wrong slot? Run either one to kick it out, then place it.

R'U'R' sexy move R'U'R' Sune R'FRF' sledgehammer gap = change grip

TWO-LOOK

OLL CROSS - yellow edges



Line bar left-to-right
F'RUR'U'f'



Hook hook pointing front-right
f'RUR'U'f'

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ 1 yellow; rest clockwise
RUR'U RU2R'



Anti-Sune $\frac{1}{2}$ 1 yellow; rest counter-clockwise
RU2R' U'RUR'



Pi $\frac{1}{2}$ 0 yellow; headlights left
f'RUR'U'f' f'RUR'U'f'

OLL CORNERS

continued



Headlights $\frac{3}{2}$ 2 yellow; headlights at the back
RD2R'U2 RD'R'U2R'



2x Headlights $\frac{3}{2}$ 2 yellow; headlights both sides
RUR'U RU'R' URU2R'



Chameleon $\frac{3}{2}$ 2 yellow next to each other
RU'R'U' FRF'



Bowtie $\frac{3}{2}$ 2 yellow diagonal
F'RUR'U' FR

FALLBACK

The badge on your row is your fallback that many Sunes, with a U turn, almost, also finishes the case. Machine-checked, there is the worst.

Top: not all yellow and nothing matches? Run Sune and read again.

No yellow edge at the top? Run the first one, then read again.

■ RUR'U sexy move ■ RUR'U Sune ■ R'FR'U' sledgehammer ■ gap = change grip

TWO-LOOK

OLL CROSS - yellow edges



Line bar left-to-right
fRUR'U'f

Hook hook pointing front-right
fRUR'U'f'

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ yellow; rest clockwise
RUR'U' RU2R'

Anti-Sune $\frac{1}{2}$ yellow; rest counter-clockwise
RU2R' U'RUR'



Pi $\frac{1}{2}$ yellow; headlights left
fRUR'U'f' fRUR'U'f'

OLL CORNERS - continued



Headlights $\frac{3}{4}$ yellow; headlights at the back
R2DR'U2' RD'R'U2R'



2x Headlights $\frac{3}{4}$ 0 yellow; headlights both sides
RUR'U' RU'R' URU2R'



Chameleon $\frac{3}{4}$ 2 yellow next to each other
rRUR'U'r' FRF'



Bowtie $\frac{3}{4}$ 2 yellow diagonal
F'rRUR'U' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked, there is the worst.

Top not all yellow and nothing matches? Run Sune and read again.

No yellow edge at all? Run the first one, then read again.

■ RUR'U' sexy move ■ RUR'U' ■ R'RF'R' sledgehammer ■ = change grip

ONE-LOOK PLI

ADJACENT CORNER SWAP

exactly one face shows headlights

T L headlights: pairs B-left F-left
RUR'U' R'F R'ZUR'UR' RUR'F'

J L solid: pairs B-right F-left R-front
x RZFRF' RU2 rUr U2

Jb L solid: pairs B-left F-back
RUR'F' RUR'U' R'F R'ZUR'U'

A L headlights: pairs F-right B-front
x LZD2 L'U'L D2 L'U'L

Ab L headlights: pairs B-right R-back
x' LZD2 LUL' D2 LU'L

F L solid
R'U'F' RUR'U' R'F R'ZUR'U' RUR'UR'

One face shows headlights, two corners of that face match. Hold as shown, turn the top face to finish.

RUR'U' sec/movs, **RUR'U'** 3 sec/movs, **R'FRF'** slide/clammer, **sec** = change dir.

ADJACENT CORNER SWAP

continued

R U L headlights: pairs F-left
Ra U'R'U' RUR DR'UR'D' R'U2R'

Rb L headlights: pairs B-left
R2F RURU'R' F'R U2R'U2R

G L headlights: pairs F-right
R2UR'UR'U' RU'R2 U'D R'UR'D'

Gb L headlights: pairs B-back
RUR'U' UD' R'ZUR'UR'U' RU'R2D

Gc L headlights: pairs B-right
RUR'U' UD' R'ZUR'UR' RU'R2 UD' R'UR'D

Gd L headlights: pairs R-front
RUR' U'D R'ZUR'UR' U'R'UR2D

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ONE-LOOK PLI

ADJACENT CORNER SWAP

exactly one face shows headlights



• T L headlights: pairs B-left F-left
RUR'U' R'F R2UR'U' RUR'F'

• J a L solid: pairs B-right F-left R-front
x R2FRF' RU2 r'Ur U2

• J b L solid: pairs B-left F-right R-back
RUR'F' RUR'U' R'F R2UR'

• A a L solid: pairs F-right B-front
x L2D2 L'U'L D2 L'U'L'

• A b L solid: pairs B-right R-back
x' L2D2 LUL' D2 LU'L

• F L solid
RU'F' RUR'U' R'F R2UR'U'

One face shows headlights: two corners of that face match. Hold as shown, and then turn the top to finish.

RUR'U' x' move **RUR'U' x' move** **R'FRF'** sliderchange **cap** = change ring

ADJACENT CORNER SWAP

continued



R a L headlights: pairs D-left R'U'R'
RU'R'U' RUR DR'UR'D' R'U2R'



R b L headlights: pairs B-left
R2F RUR'U' FR U2RU2R



G a L headlights: pairs F-right
R2UR'U'R' RU'R2 U'D R'UR'D'



G b L headlights: pairs R-back
RUR'U' UD' R2UR'U'R' RU'R2R'



G c L headlights: pairs B-right
R2UR'U'R' RU'R2 UD' RU'R'D'



G d L headlights: pairs R-front
RUR'U' UD' R2UR'U'R' RU'R2R'

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ANNEX – BIG CUBES AND THE MAP

BIG CUBES: 4x4 and 5x5

Reduce: build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY – never widen it.

Last 2 edges

4x4 PLL parity

4x4 OLL parity = 1 edge pair flipped - half the time

Rw U2 x Rw U2 Rw U2 Rw' U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw'

5x5 edges parity = 1 edge pair flipped - half the time + one move differs; 5x5 has no PLL parity!

Rw U2 x Rw U2 Rw U2 **3Rw** U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw'

OLL parity = 1 edge pair before OLL and PLL – both parity signs

NOTATION

R U F D B = turn that face clockwise, from outside the cube. = "rotate around the axis that turns the cube".

M = middle slice, turning the way I turns. x y z = the whole cube, turning the way R turns. i = inner, p = piecewise r f = that face plus the layer behind it, turning together.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm that you hands runs at one unit - sludge move = the R U2 U' trigger - permutation = the R-F-R' trigger, parity = a case a 3x3 cannot have - 2 edges glued into one, on a 4x4 or 5x5

RURU = setup move **RURU** SU = **FRF** sledhammer gap = change grip

ANNEX — BIG CUBES AND THE MAP

BIG CUBES — 4x4 and 5x5

Reduce the 6 centers — pair the 12 edges — solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY — never widen it.

Last 2 edges

4x4 PLL parity

4x4 OLL parity = 1 edge pair flipped - half the time

Rw U2 x Rw U2 Rw U2 Rw' U2 Lw U2 Lw' U2 Rw U2 Rw' U2 Rw' U2

5x5 edge parity = 1 edge pair flipped - half the time - *one more* downface, 5x5 has no PLL parity

Rw U2 x Rw U2 Rw U2 **3Rw** U2 Lw U2 Lw' U2 Rw U2 Rw' U2 Rw' U2

NOTATION

R U F L B = turn that face clockwise, from outside the cube. = "

turning the 4x4 and 5x5 layers. OLL and PLL — both parity algorithms

M = middle slice, turning the way I turns. x = the whole cube, turning the way R turns. i = 180 degrees, overcase r l = a face plus the layer behind it, turning together.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm you hands run as one unit - *slay* move = into one unit - R U F or trigger -

RURU = slice move **RURU** = slice **R'RF** = slicechange **grip** = change grip

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the SHORT edge. The numeral on both faces is your check: cut ONE card first. This bar must measure exactly 80 mm:

[illegible]

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